

Group members' name and student ID:

Assignment group number:

BMDS2003 DATA SCIENCE

CLO1 - Apply appropriate data science concepts, statistics, and machine learning to make predictions on data. (C3, PLO2)

Criterion	Missing / Unacceptable	Poor	Good	Excellent	Marks
Model Selection (CLO1)	No justification for model selection. Approach irrelevant or lacks evidence. <i>0–5 marks</i>	Limited or weak justifications for model selection. Choices appear arbitrary or poorly supported. <i>6–10 marks</i>	Adequate justifications are presented. The chosen model is generally appropriate and supported by reasonable evidence or rationale. Explanation of algorithm behaviour and comparison with alternatives is acceptable. <i>11–17 marks</i>	Strong and insightful justification for model selection. The model is highly suitable and well-supported by evidence. Student demonstrates a deep understanding of algorithm suitability and optimisation. Advanced or creative configurations (e.g., using variants with fine-tuned architectures) are evident and aligned with predictive goals. <i>18–25 marks</i>	

CLO2 – Practice exploratory data analysis and visualisation methods with appropriate data science tools. (P3, PLO3)

Criterion	Missing / Unacceptable	Poor	Good	Excellent	Marks
Data Preprocessing (CLO2)	Data preprocessing steps are missing or irrelevant. <i>0–2 marks</i>	Data preprocessing steps are incomplete and not described clearly. <i>3–5 marks</i>	Relevant preprocessing steps are explained, offering clear details about the components. <i>6–7 marks</i>	The preprocessing steps are informative, succinct, and provide detailed insights into the data science components. <i>8–10 marks</i>	
Descriptive and Exploratory Data Analysis (CLO2)	Descriptive and exploratory analytics are missing or completely irrelevant. No meaningful analysis or interpretation of data patterns. <i>0–4 marks</i>	Analytical explanations are minimal, unclear, or poorly aligned with data science and business intelligence objectives. Limited understanding of descriptive or exploratory analysis is shown. <i>5–8 marks</i>	Descriptive and exploratory analytics are appropriately conducted with clear explanations. Visuals and summaries reveal relevant insights supported by logical reasoning and proper data interpretation. <i>9–12 marks</i>	Comprehensive and insightful descriptive and exploratory analytics are performed. Analyses are well-structured, supported by meaningful visualisations, and demonstrate a deep understanding of data relationships and business intelligence concepts. Evidence of thoughtful interpretation, trend discovery, or correlation analysis adds value. <i>13–15 marks</i>	
Graphing and Visualisation (CLO2)	Visualisations missing or irrelevant. <i>0–2 marks</i>	Limited or unclear visualisations that confuse information. <i>3–4 marks</i>	Clear graphs and visualisations are presented. <i>5–7 marks</i>	Clear visualisations presented using appropriate Python tools (e.g., Matplotlib, Seaborn, Plotly). <i>8–10 marks</i>	

CLO3 – Analyse relevant techniques, statistical methods, and machine learning algorithms for data analytics. (C4, PLO7)

Criterion	Missing / Unacceptable	Poor	Good	Excellent	Marks
Advanced Analytics and Discussion (CLO3)	No advanced analytics presented. Results are missing, irrelevant, or copied. <i>0–7 marks</i>	Basic or incomplete analysis. Limited discussion of results, with minimal interpretation or weak connection to prediction goals. Prototype lacks functionality or clarity. <i>8–15 marks</i>	Appropriate analytics are conducted using relevant machine learning models. Comparisons and evaluations are clear and supported with reasoning. Prototype is simple but functional, showing basic integration of results. <i>16–22 marks</i>	Comprehensive and critical analysis. Models are effectively compared and justified with literature or evidence. Clear interpretation of results, limitations, and improvements discussed. Includes a functional prototype demonstrating real-world deployment or interactive results. <i>23–30 marks</i>	
Report Structure (CLO3)	Report is incomprehensible, irrelevant, or poorly structured. <i>0–1 mark</i>	Structure is weak and disorganised, making the report difficult to follow. <i>2 marks</i>	Report structure is adequate, with a clear flow and reasonable organisation. <i>3–4 marks</i>	Report structure is excellent, with smooth transitions, logical flow, and comprehensive organisation aligned with data science concepts. <i>5 marks</i>	
Presentation (CLO3)	No presentation delivered, or presentation is incoherent and unprepared. <i>0–1 mark</i>	Presentation is disorganised with minimal preparation. Team shows weak understanding or coordination. Attire and delivery lack professionalism. <i>2 marks</i>	Presentation is clear and reasonably structured. All members participate and show understanding. Attire and delivery are appropriate with minor issues. <i>3–4 marks</i>	Well-prepared and confident presentation with smooth coordination among members. Demonstrates clear understanding, professional attire, and strong delivery. Demo effectively supports the explanation. <i>5 marks</i>	

Remark:

TOTAL

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